

CM0832 Elements of Set Theory

3. Relations and Functions

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Semester 2017-2

Preliminaries

Convention

The number assigned to chapters, examples, exercises, figures, sections, and theorems on these slides correspond to the numbers assigned in the textbook [Enderton 1977].

Ordered Pairs

Observation

Let a and b be sets. An ordered pair $\langle a, b \rangle$ should be a **set** such that

$$\langle a, b \rangle = \langle c, d \rangle \quad \text{iff} \quad a = c \wedge b = d.$$

Definition

We define an **ordered pair** using Kuratowski's definition, that is,

$$\langle a, b \rangle := \{\{a\}, \{a, b\}\}.$$

Ordered Pairs

Example

We show that $\langle \emptyset, \{\emptyset\} \rangle \neq \langle \{\emptyset\}, \emptyset \rangle$.

$$\begin{aligned}\langle \emptyset, \{\emptyset\} \rangle &= \{\{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\ &= \{\{\emptyset\}, \{\{\emptyset\}, \emptyset\}\} \\ &\neq \{\{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\}\} \\ &= \langle \{\emptyset\}, \emptyset \rangle.\end{aligned}$$

Ordered Pairs

Example

Let a be a set. Then

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Exercise

To give a different definition of ordered pair.

Cartesian Product

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Let A and B be sets. The **Cartesian product** of A and B is defined by

$$A \times B := \{ \langle x, y \rangle \mid x \in A \wedge y \in B \}.$$

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Let A and B be sets. Note that $A \times B$ is a set because we can define it via the subset axiom scheme.

$$A \times B := \{ \langle x, y \rangle \in \mathcal{PP}(A \cup B) \mid x \in A \wedge y \in B \}.$$

Relations

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A **relation** is a set of ordered pairs.

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Let R the relation defined by $R = \{\langle a, b \rangle, \langle b, b \rangle, \langle c, b \rangle\}$. Diagram: whiteboard.

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Let R the relation defined by $R = \{\langle a, b \rangle, \langle b, b \rangle, \langle c, b \rangle\}$. Diagram: whiteboard.

Example

Let $\omega = \{0, 1, 2, \dots\}$. The identity relation on ω is defined by

$$\begin{aligned} I_\omega &:= \{ \langle n, n \rangle \mid n \in \omega \} \\ &= \{ \langle 0, 0 \rangle, \langle 1, 1 \rangle, \langle 2, 2 \rangle, \dots \}. \end{aligned}$$

Relations

Definition

Let R be a relation. We define the **domain**, the **range** and the **field** of R by

$$\text{dom } R := \{ x \mid \exists y (\langle x, y \rangle \in R) \},$$

$$\text{ran } R := \{ y \mid \exists x (\langle x, y \rangle \in R) \},$$

$$\text{fld } R := \text{dom } R \cup \text{ran } R.$$

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Observation

Let R be a relation. Note that $\text{dom } R$ and $\text{ran } R$ are sets because we can define them via the subset axiom scheme.

$$\text{dom } R := \left\{ x \in \bigcup \bigcup R \mid \exists y (\langle x, y \rangle \in R) \right\},$$

$$\text{ran } R := \left\{ y \in \bigcup \bigcup R \mid \exists x (\langle x, y \rangle \in R) \right\}.$$

n -Ary Relations

Definition

We define an **ordered n -tuple**, for $n \geq 3$, by

$$\langle x_1, x_2, \dots, x_n \rangle := \langle \langle x_1, x_2, \dots, x_{n-1} \rangle, x_n \rangle$$

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Example

Ordered triple (**3**-tuple) and ordered quadruple (**4**-tuple).

$$\begin{aligned}\langle x_1, x_2, x_3 \rangle &:= \langle \langle x_1, x_2 \rangle, x_3 \rangle, \\ \langle x_1, x_2, x_3, x_4 \rangle &:= \langle \langle x_1, x_2, x_3 \rangle, x_4 \rangle.\end{aligned}$$

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Example

Ordered triple (**3-tuple**) and ordered quadruple (**4-tuple**).

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Definition

We define an **1-tuple** by

$$\langle x \rangle := x.$$

n -Ary Relations

Definition

Let A be a set. We define an n -ary relation on A to be a set of ordered n -tuples with all components in A .

n -Ary Relations

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Example

Whiteboard.

Observation

Let A be a set. Note that an 1-ary relation on A is just a subset of A but it is not a relation.

Functions

Definition

A **function** (**mapping** or **correspondence**) is a **relation** F such that for each x in $\text{dom } F$ there is only one y such that xFy .

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Notation

We write $F : A \rightarrow B$ iff F is a function, $\text{dom } F = A$ and $\text{ran } F \subseteq B$.

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Notation

We write $F : A \rightarrow B$ iff F is a function, $\text{dom } F = A$ and $\text{ran } F \subseteq B$.

Definition

Let F be a function and A and B sets.

- (i) F is a function **on** (**from**) A iff $\text{dom } F = A$.
- (ii) F is a function **into** (**to**) B iff $\text{ran } F \subseteq B$.
- (iii) F is a function **onto** B iff $\text{ran } F = B$.

Functions

Exercise 3.11

Prove the following version (for functions) of the extensionality principle: Assume that F and G are functions, $\text{dom } F = \text{dom } G$, and $F(x) = G(x)$ for all x in the common domain. Then $F = G$.

Functions

Definition

A function F is **one-to-one** (or **injective**) iff for each $y \in \text{ran } F$ there is only one x such that xFy . In other words, if $x_1, x_2 \in \text{dom } F$ and $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.

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Functions

Definition

Let A, F and G be **sets**. We define, the **inverse** of F , the **composition** of F and G , the **restriction** of F to A and the **image** of A under F by

$$F^{-1} := \{ \langle y, x \rangle \mid xFy \} \quad (\text{inverse of } F)$$

$$F \circ G := \{ \langle x, y \rangle \mid \exists t (xGt \wedge tFy) \} \quad (\text{composition of } F \text{ and } G)$$

$$F \upharpoonright A := \{ \langle x, y \rangle \mid x \in A \wedge xFy \} \quad (\text{restriction of } F \text{ to } A)$$

$$\begin{aligned} F[A] &:= \text{ran}(F \upharpoonright A) \\ &= \{ y \mid \exists x (x \in A \wedge xFy) \} \end{aligned} \quad (\text{image of } A \text{ under } F)$$

Functions

Example

Let

$$F = \{\langle \emptyset, a \rangle, \langle \{\emptyset\}, b \rangle\}.$$

Then

$$\text{dom } F = \{\emptyset, \{\emptyset\}\}$$

$$\text{ran } F = \{a, b\},$$

$$F^{-1} = \{\langle a, \emptyset \rangle, \langle b, \{\emptyset\} \rangle\},$$

$$F \upharpoonright \emptyset = \emptyset,$$

$$F \upharpoonright \{\emptyset\} = \{\langle \emptyset, a \rangle\},$$

$$F[\{\emptyset\}] = \{a\},$$

$$F(\{\emptyset\}) = b.$$

F is a function,

F^{-1} is function iff $a \neq b$,

Functions

Exercise 3.18

Let R be the set

$$\{\langle 0, 1 \rangle, \langle 0, 2 \rangle, \langle 0, 3 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 3 \rangle\}.$$

To find $R \circ R$, $R \upharpoonright \{1\}$, $R^{-1} \upharpoonright \{1\}$, $R[\{1\}]$ and $R^{-1}[\{1\}]$.

Functions

Exercise 3.18

Let R be the set

$$\{\langle 0, 1 \rangle, \langle 0, 2 \rangle, \langle 0, 3 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 3 \rangle\}.$$

To find $R \circ R$, $R \restriction \{1\}$, $R^{-1} \restriction \{1\}$, $R[\{1\}]$ and $R^{-1}[\{1\}]$.

Exercise (p. 44)

Let A, F and G be sets. Show that F^{-1} , $F \circ G$, $F \restriction A$ and $F[A]$ are sets.

Functions

Theorem 3E

Let F be a set. Then

$$\text{dom } F^{-1} = \text{ran } F \quad \text{and} \quad \text{ran } F^{-1} = \text{dom } F.$$

If additionally F is a relation, then

$$(F^{-1})^{-1} = F.$$

Functions

Theorem 3G

Let F be an one-to-one function.

- If $x \in \text{dom } F$, then

$$F^{-1}(F(x)) = x.$$

- If $y \in \text{ran } F$, then

$$F(F^{-1}(y)) = y.$$

Functions

Theorem 3H

Let F and G be functions. Then

- $F \circ G$ is a function,
- $\text{dom}(F \circ G) = \{x \in \text{dom } G \mid G(x) \in \text{dom } F\}$ and
- if $x \in \text{dom}(F \circ G)$, then $(F \circ G)(x) = F(G(x))$.

Functions

Theorem 3I

Let F and G be sets. Then

$$(F \circ G)^{-1} = G^{-1} \circ F^{-1}.$$

Functions

Theorem 3J

Let F be a function $F : A \rightarrow B$ and $A \neq \emptyset$.

- (i) There exists a function $G : B \rightarrow A$ (a “left inverse”) such that $G \circ F$ is the identity function I_A on A iff the function F is one-to-one.
- (ii) There exists a function $H : B \rightarrow A$ (a “right inverse”) such that $F \circ H$ is the identity function I_B on B iff the function F maps A onto B .

Functions

Axiom of choice (first form)

For any relation R there is a function $H \subseteq R$ with $\text{dom } H = \text{dom } R$.

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Example

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Observation

Is the axiom of choice accepted in constructive mathematics? (See, e.g. Martin-Löf [2006]).

Functions

Definition

Let A and B be sets. We define the **set of functions** from A into B by

$$B^A := \{ F \mid F : A \rightarrow B \} =: {}^A B.$$

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Example

- $\{0, 1\}^\omega$: The set of infinity binary sequences.

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- $\emptyset^A = \emptyset$ for $A \neq \emptyset$ (no function can have a non-empty domain and an empty range).

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Example

- $\{0, 1\}^\omega$: The set of infinity binary sequences.
- $\emptyset^A = \emptyset$ for $A \neq \emptyset$ (no function can have a non-empty domain and an empty range).
- $A^\emptyset = \{\emptyset\}$ for any set A ($\emptyset$ is the only function with an empty domain).

Functions

Observation

Let A and B be sets. Note that B^A is a set because we can define it via the subset axiom scheme.

$$B^A := \{ F \in \mathcal{P}(A \times B) \mid F : A \rightarrow B \}.$$

Families

Observation

Families is another way to express functions when the range of a function is more important than the function itself. We write functions as families when we want to put the emphasis on the values of the function rather in the function.[†]

[†]Enderton [1977] do not use families, but 'only' functions.

Families

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Families is another way to express functions when the range of a function is more important than the function itself. We write functions as families when we want to put the emphasis on the values of the function rather in the function.[†]

Observation

The terminology and notation on families is not established.

[†]Enderton [1977] do not use families, but 'only' functions.

Families

Definition

Let I and X be sets. A **family in X indexed by I** is a function

$$A : I \rightarrow X$$

$$A = \{ \langle i, A_i \rangle \mid i \in I \text{ and } A_i \in X \},$$

where $A_i := A(i)$, for all $i \in I$.[†] The set I is the **index set** of the family.

[†]See, e.g. Halmos [1960], Drake [1974] and Hamilton [(1982) 1992].

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Notation

The above family A is denoted by $\langle A_i \mid i \in I \rangle$ following to [Hrbacek and Jech (1978) 1999].

[†]See, e.g. Halmos [1960], Drake [1974] and Hamilton [(1982) 1992].

Families

Definition

The **union** of a family $\langle A_i \mid i \in I \rangle$ is defined by

$$\begin{aligned}\bigcup_{i \in I} A_i &:= \bigcup \{ A_i \mid i \in I \} \\ &= \{ x \mid x \in A_i \text{ for some } i \text{ in } I \}.\end{aligned}$$

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Example

Whiteboard.

Families

Definition

The **intersection** of a family $\langle A_i \mid i \in I \rangle$ is defined by

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Example

Whiteboard.

Families

Definition

The **Cartesian product** (or **generalised product**) of a family $\langle A_i \mid i \in I \rangle$ is defined by

$$\prod_{i \in I} A_i := \left\{ f \mid f : I \rightarrow \bigcup_{i \in I} A_i \text{ and } \forall i (i \in I \rightarrow f(i) \in A_i) \right\} =: \prod_{i \in I} A_i.$$

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Example

Let $\langle A_i \mid i \in I \rangle$ be a family. If $A_i = B$ for all $i \in I$, then

$$\begin{aligned} \prod_{i \in I} A_i &= B^I \\ &= \{ f \mid f : I \rightarrow B \}. \end{aligned}$$

Families

Example

The following example illustrates the generalisation of the Cartesian product.

Let X and Y be two sets. Recall that the Cartesian product of X and Y was defined by

$$X \times Y := \{ \langle x, y \rangle \mid x \in X \wedge y \in Y \}.$$

(continued on next slide)

Families

Example (continuation)

Let $I = \{a, b\}$ be an index set and let $\langle Z_i \mid i \in I \rangle$ be a family where $Z_a = X$ and $Z_b = Y$. Then

$$\bigtimes_{i \in I} Z_i = \{ f \mid f : I \rightarrow X \cup Y, \text{ such that } f(a) \in X \text{ and } f(b) \in Y \}.$$

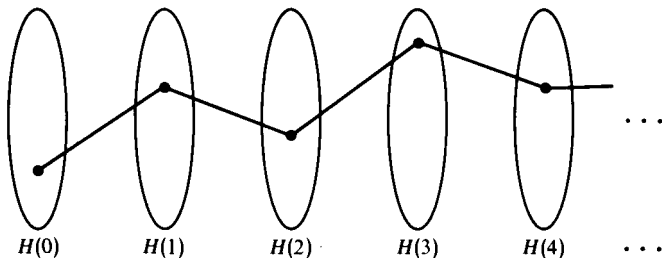
Now, we can define the one-to-one correspondence

$$\begin{aligned} h : \bigtimes_{i \in I} Z_i &\rightarrow X \times Y \\ h(f) &= \langle f(a), f(b) \rangle. \end{aligned}$$

Families

Axiom of choice (second form)

Let $\langle H_i \mid i \in I \rangle$ be a family. If $H(i) \neq \emptyset$ for all $i \in I$, then $\prod_{i \in I} H(i) \neq \emptyset$.[†]



[†]Figure source: Enderton [1977, Fig. 11]

Equivalence Relations

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- **reflexive** iff xRx for all $x \in A$,
- **symmetric** iff xRy implies yRx for all $x, y \in A$ and
- **transitive** iff xRy and yRz imply xRz for all $x, y, z \in A$.

Example

Whiteboard.

Equivalence Relations

Introduction

Whiteboard.

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Questions

- Let $A = \{a, e, i, o, u\}$. Is the equality relation on A an equivalence relation?

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Equivalence Relations

Questions

- Let $A = \{a, e, i, o, u\}$. Is the equality relation on A an equivalence relation?
- Let $A \neq \emptyset$ be a set. Is the relation $\emptyset$ on A an equivalence relation?
- Let A be a set. Is the relation $A \times A$ an equivalence relations?
- Let A be a singleton. It is possible to define an equivalence relation on A ?

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Definition

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Definition

Let R be an equivalence relation on a set A and let $x \in \text{fld } R$. The set $[x]_R$ is the **equivalence class of x (modulo R)**.

Notation

We write $[x]$ if the relation R is clear in the context.

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Example

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Equivalence Relations

Theorem 3N

Let R be an equivalence relation on a set A and let $x, y \in A$. Then

$$[x]_R = [y]_R \quad \text{iff} \quad xRy.$$

Equivalence Relations

Theorem 3P

Let R be an equivalence relation on a set A . Then the set

$$\{ [x]_R \mid x \in A \}$$

of all equivalence classes is a partition of the set A .

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Exercise 3.37

Assume that Π is a partition of a set A . Define the relation R_Π as follows:

$$xR_\Pi y \quad \text{iff} \quad (\exists B \in \Pi)(x \in B \wedge y \in B).$$

Show that R_Π is an equivalence relation on A .

Equivalence Relations

Definition

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$$A/R := \{ [x]_R \mid x \in A \}.$$

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Observation

Using the λ -notation we could define the natural map by the anonymous function $\lambda x.[x]_R$.

Linear Ordering Relations

Motivation

What means that R is an ordering relation on a set A ?

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What means that R is an ordering relation on a set A ?

Definition

Let R be a binary relation on a set A . The relation R satisfies **trichotomy** if exactly one of the three alternatives

$$xRy, \quad x = y \quad \text{or} \quad yRx$$

holds for all $x, y \in A$.

Linear Ordering Relations

Definition

Let A be a set. A **linear ordering** (or **total ordering**) on A is a binary relation R on A such that:

- (i) R is transitive relation and
- (ii) R satisfies trichotomy.

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Definition

Let A be a set. A **linear ordering** (or **total ordering**) on A is a binary relation R on A such that:

- (i) R is transitive relation and
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Example



References

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